Alessandro Guenna

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alessandroguenna.github.io

I am a young mechanical and energy engineer with a strong passion for the development of innovative solutions in the context of the green transition. Confident in the use of Python and MATLAB, which I practice in personal projects showcased on my portfolio website. Seeking roles in energy systems modelling, gas turbine performance, process engineering for energy efficiency and heat recovery, and techno-economic analysis of energy plants. **Available from December 2026.**

Experience

Mälardalen University (MDU), Västerås, Sweden - Master's thesis internship — Aug 2026 – present

- Techno-economic study on waste heat recovery in the steel industry
- Developing a MATLAB model for different plant layouts based on Organic Rankine Cycle and District Heating, simulating design and off-design performance
- Assessing energy recovery, payback period, LCOE and CO_2 savings

Education

University of Genoa, Genoa, Italy - Master's degree in Energy Engineering — 2024 – Dec 2026 (expected)

- Current average: 110/110

University of Genoa, Genoa, Italy - Bachelor's degree in Mechanical Engineering — 2020 – 2024

- Final grade 103/110
- Thesis: experimental evaluation of the performance of a magnetic refrigerator prototype, in collaboration with the company Bi-Freezer s.r.l.

University of Seville, Seville, Spain - Erasmus BIP "Energy communities for smart cities" — February 2026

- Technical analysis for optimal sizing of PV plants based on load demand
- Business and regulatory assessment of REC projects

Projects

Microturbine control panel (CHP) - MATLAB, Simulink, App Designer — Project page

- Developed a real-time dynamic model of an ∼80 kW recuperated gas microturbine for cogeneration, including compressor map, choked turbine, 5-node recuperator with thermal inertia and shaft dynamics
- Designed a PI fuel control system with turbine-inlet temperature limits and an optimal-speed schedule maximising electrical efficiency within surge, choking and speed constraints
- Built an interactive control panel app to set the load and monitor the plant response in real time

Winner of "Energy and Innovation – From Knowledge to Entrepreneurship", organized by Wylab and TICASS — 2025

- Designed a business model for an industrial plant converting plastic waste into liquid fuel via catalytic pyrolysis (suitable for Otto cycle engines), promoting circular economy

Technical skills

Programming languages: Python, Excel VBA for automated worksheets

Software: MATLAB and Simulink, Ansys Fluent and CFX, Autodesk Inventor, HOMER Pro, Microsoft Office

Languages

Italian (native), English (advanced), German (basic)